# Circuits III

Thevenin's Theorem

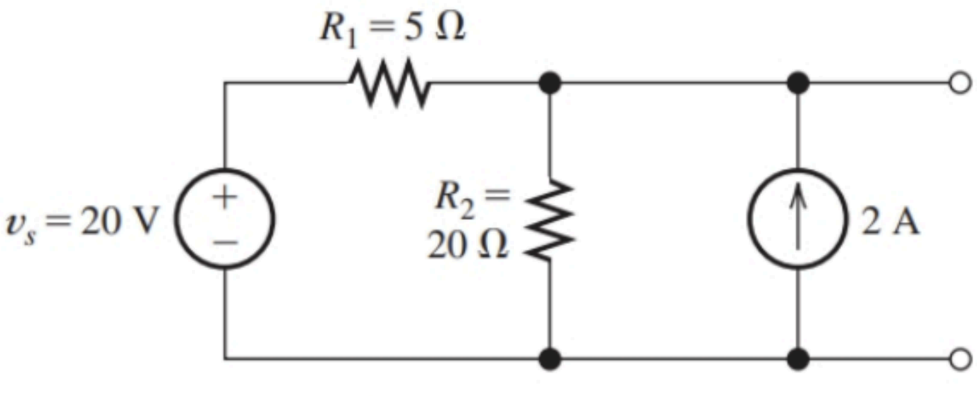
Thevenin's Theorem states that an open circuit of a combination of different power sources and resistances can be represented as a equivalent series branch of a single voltage source and resistance. The voltage value is labelled as the Thevenin voltage while the resistance is labelled as the Thevenin resistance.

Usually, the voltage can be calculated normally like a usual circuit problem. However, the resistance can be found with 2 methods, either by determining the short circuit current and using ohms law, or if only independent voltage sources are present, voltage sources can be shorted and current sources can be opened for the effective resistance across the terminal to be determined as the Thevenin resistance.

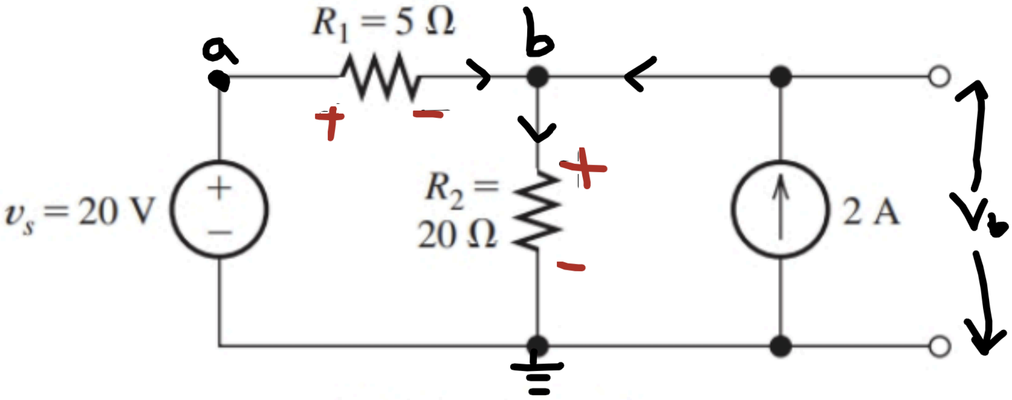
Experimentally, voltage and current can be found with a multimeter, and the experimental value is found.

Thevenin's Equivalent on Independent Power Sources

Say we had to determine the Thevenin voltage and resistance $(V_{th}, R_{th})$ of the circuit:




We can use node voltage analysis at point b to find the open circuit voltage, which is the Thevenin voltage:



(a) Original circuit

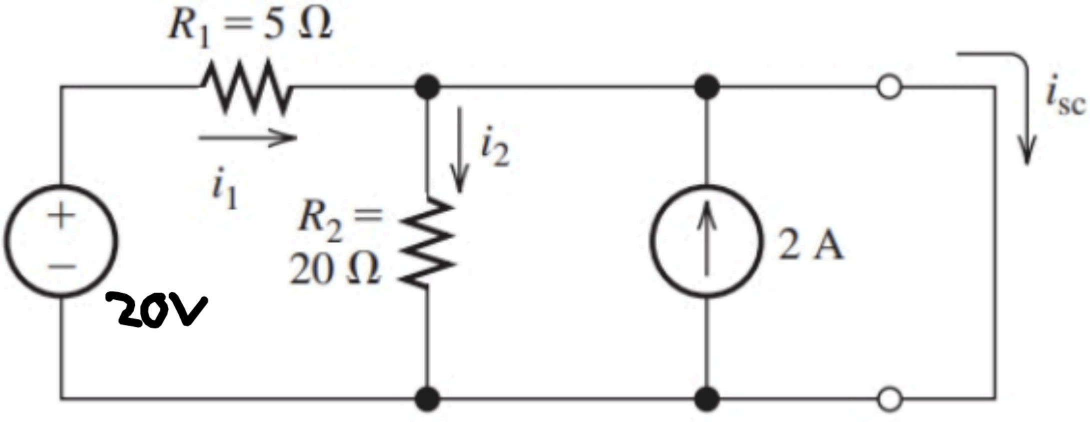
Thus,

$$v_a = 20V$$
$$v_b : \frac{v_a - v_b}{5} + 2 - \frac{v_b}{20} = 0$$

Hence,

$$\frac{20 - v_b}{5} + 2 - \frac{v_b}{20} = 0$$
$$4(20 - v_b) + 2(20) - v_b = 0$$
$$80 - 4v_b + 40 - v_b = 0$$
$$120 - 5v_b = 0$$
$$v_b = \frac{120}{5} = 24V$$
$$\therefore V_{th} = 24V$$

Then, the Thevenin resistance can be found by first finding the short circuit current.



Notice how R2 is shorted, thus current does not flow through R2.

Using the node above the 2A source, we can apply KCL to find that

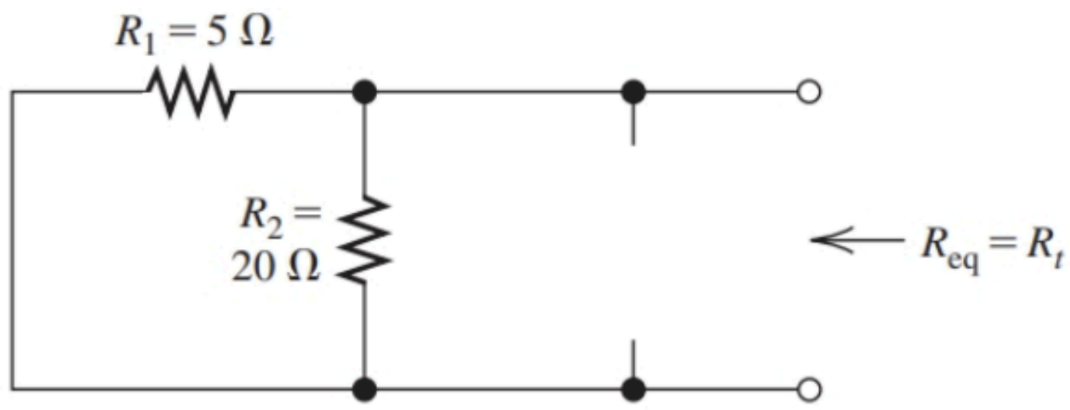
$$\begin{aligned} i_{sc} &= \frac{20}{5} + 2 \\ &= 6A \end{aligned}$$

And then, the Thevenin resistance can be determined as

$$R_{th} = \frac{V_{th}}{i_{sc}} = \frac{24}{6} = 4\Omega$$

Alternatively, we could determine Thevenin's resistance by using equivalent resistance formulas, as the circuit only has independent power sources.

By shorting voltage sources and opening current sources, we get



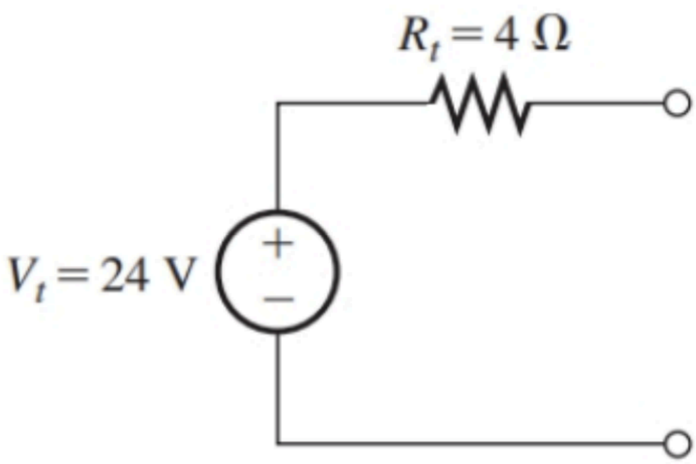

And hence the Thevenin's resistance can be determined as

$$R_{th} = \frac{5 \times 20}{5 + 20} = 4\Omega$$

The Thevenin's resistance may also be found first via this method, to then determine the Thevenin's voltage:

$$\begin{aligned} i_{sc} &= 6A \text{ (same method as earlier)} \\ \therefore V_{th} &= i_{sc} R_{th} \\ &= 6 \times 4 = 24V \end{aligned}$$

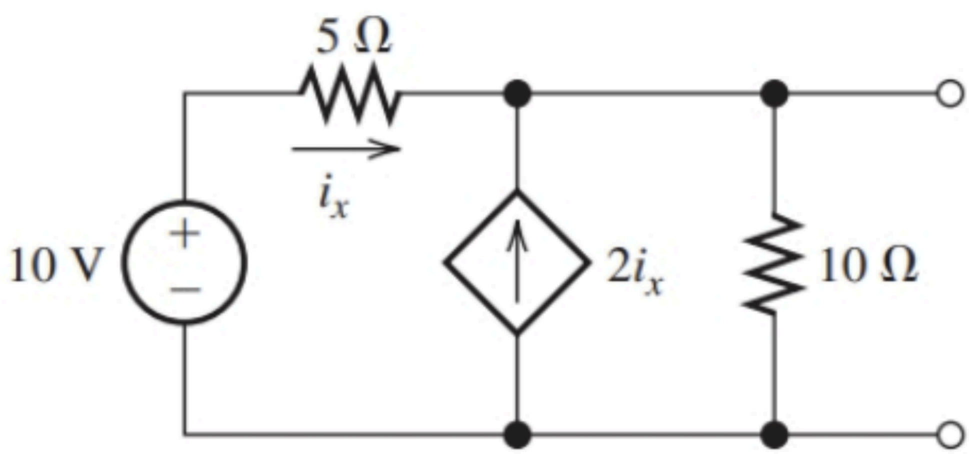
Either way, the circuit can now be represented as a voltage source and resistor in series as



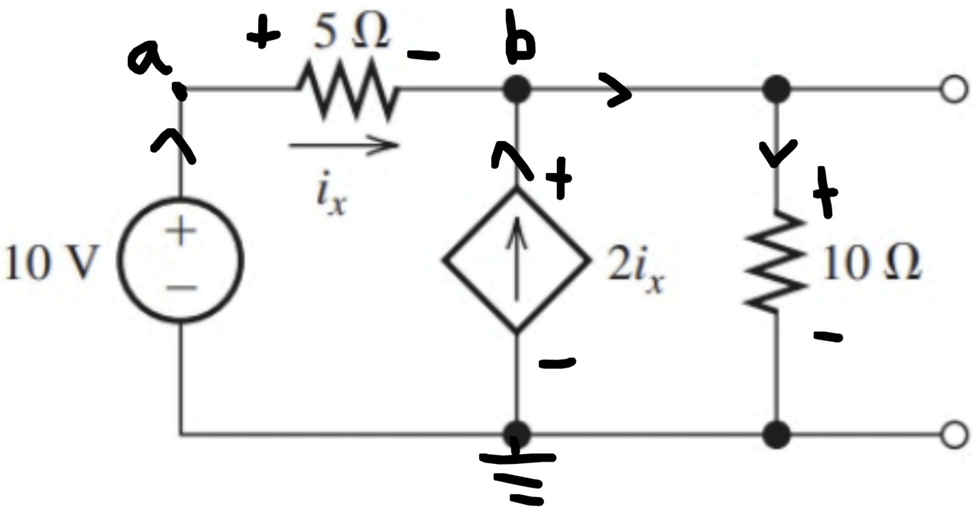

Thevenin’s Equivalent on Dependent Power Sources

Zeroing the sources only worked when power sources were independent. For dependent power sources, we must use standard analysis methods, like node voltage analysis.

Say we had to find the Thevenin’s Equivalent circuit of the following:




We can assign nodes and currents like so:




and hence the open circuit voltage is vb. Note that the right adjacent node of b has the same voltage as vb, thus we must consider the current flowing through both nodes together. Thus,

$$v_a = 10V$$
$$v_b : i_x + 2i_x - \frac{v_b}{10} = 0$$
$$3i_x - \frac{v_b}{10} = 0$$

We know that

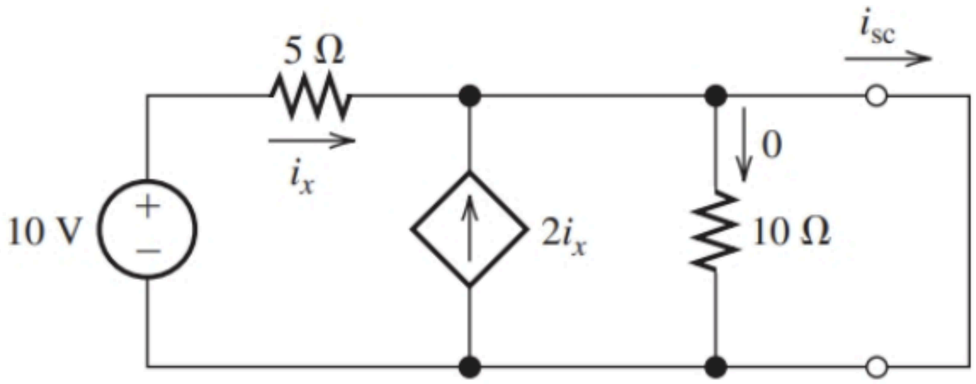
$$i_x = \frac{v_a - v_b}{5}$$

and thus

$$\frac{3(v_a - v_b)}{5} - \frac{v_b}{10} = 0$$
$$6(v_a - v_b) - v_b = 0$$
$$6(10 - v_b) - v_b = 0$$
$$60 - 6v_b - v_b = 0$$
$$60 - 7v_b = 0$$
$$v_b = \frac{60}{7} = 8.57V$$

Thus, $V_{th} = 8.57V$.

then, the short circuit current can be found by using KCL on node b:




$$i_{sc} = i_x + 2i_x = 3i_x$$

ix can be found via ohms law on the 5 ohm resistor:

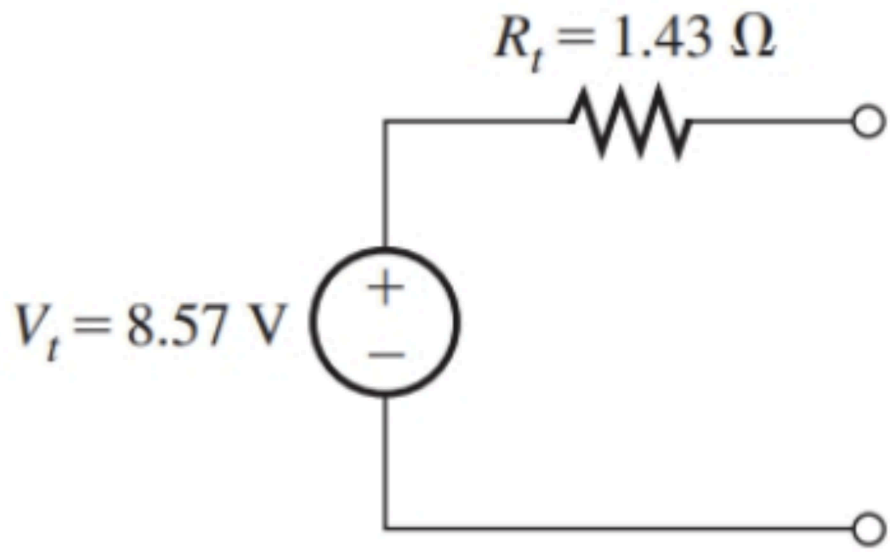
$$i_x = \frac{10}{5} = 2A$$
$$\therefore i_{sc} = 3(2) = 6A$$

Note that the previous formula of $i_x = \frac{v_a - v_b}{5}$ with v_b as 8.57V cannot be used as shorting changes the voltage of vb.

With the short circuit current and the Thevenin's voltage, the Thevenin's resistance can be found as

$$R_{th} = \frac{V_{th}}{i_{sc}} = \frac{8.57}{6} = 1.43\Omega$$

Hence, the Thevenin's equivalent circuit is

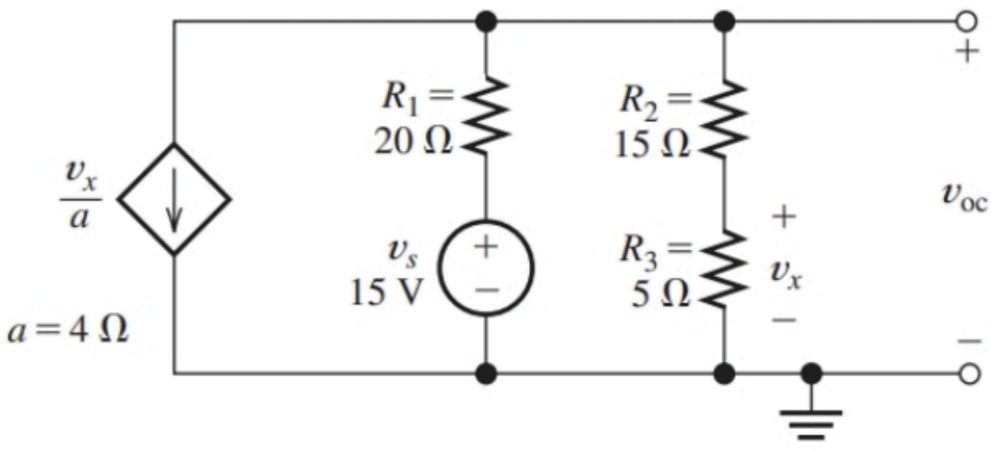



Norton's Theorem

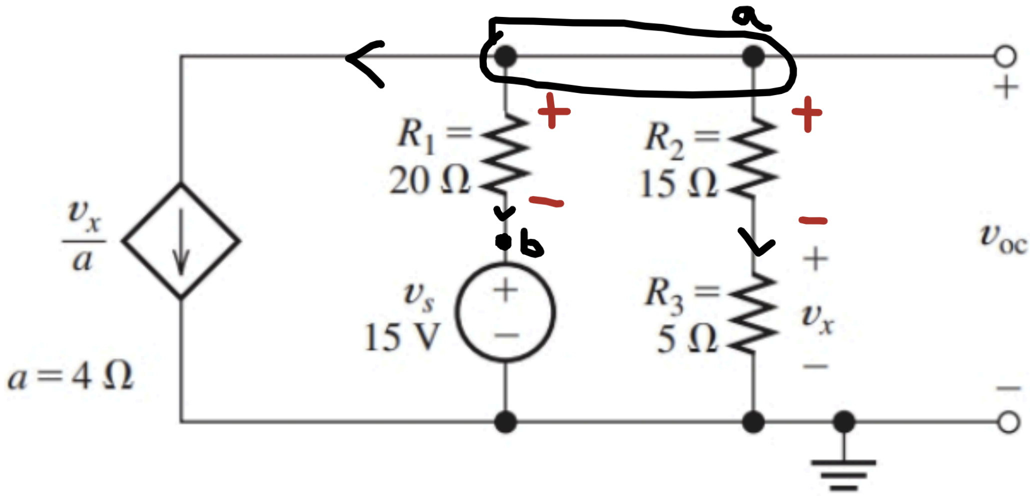
Norton's Theorem is a similar theorem to Thevenin's theorem, where instead any open circuit can be represented by a current source and resistance in parallel.

The current source can be found by shorting the terminals, hence $I_n = i_{sc}$. The resistance is actually the same as the Thevenin's resistance, hence it can be determined with similar methods.

Say we had to determine the Norton's Equivalent Circuit of the following:




Again, define our currents and voltages:




We can let the upper node be a, and we must group these two nodes as a as they have the same potential. We also set a node b, such that $v_b = 15V$.

Hence, find the KCL at node a:

$$-\frac{v_x}{4} - \frac{v_a - v_b}{20} - \frac{v_a}{15+5} = 0$$
$$-\frac{v_x}{4} - \frac{v_a - 15}{20} - \frac{v_a}{20} = 0$$
$$-5(v_x) - (v_a - 15) - v_a = 0$$
$$-5v_x - 2v_a + 15 = 0$$

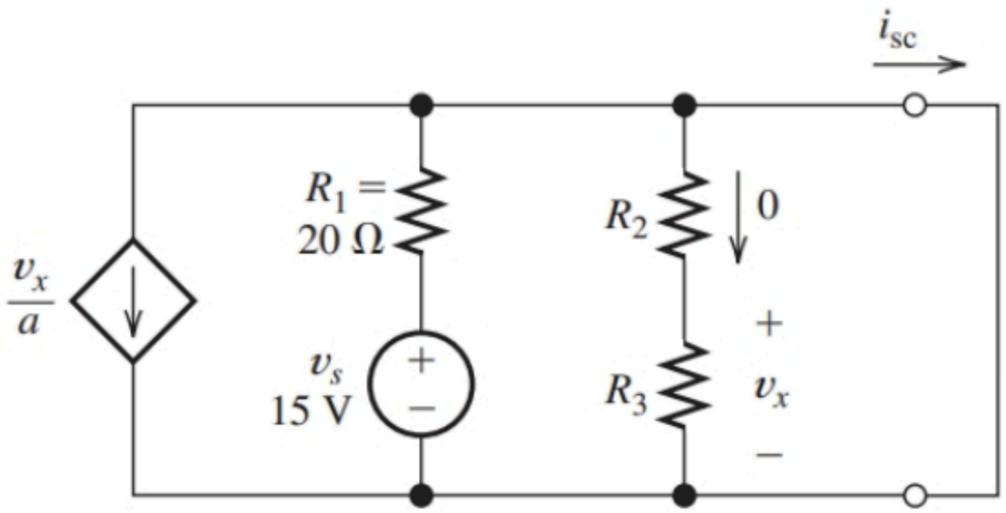
We can also determine vx in terms of va, via VDR, as

$$v_x = \frac{5}{5+15} \times v_a$$
$$v_x = \frac{v_a}{4}$$

Thus,

$$-\frac{5}{4}v_a - 2v_a + 15 = 0$$
$$-3.25v_a + 15 = 0$$
$$v_a = \frac{15}{3.25}V = \frac{60}{13}V = 4.62V$$

For the short circuit current, we can realise that R2 and R3 are shorted




Hence, vx = 0 and thus the dependent current source will be 0 too.

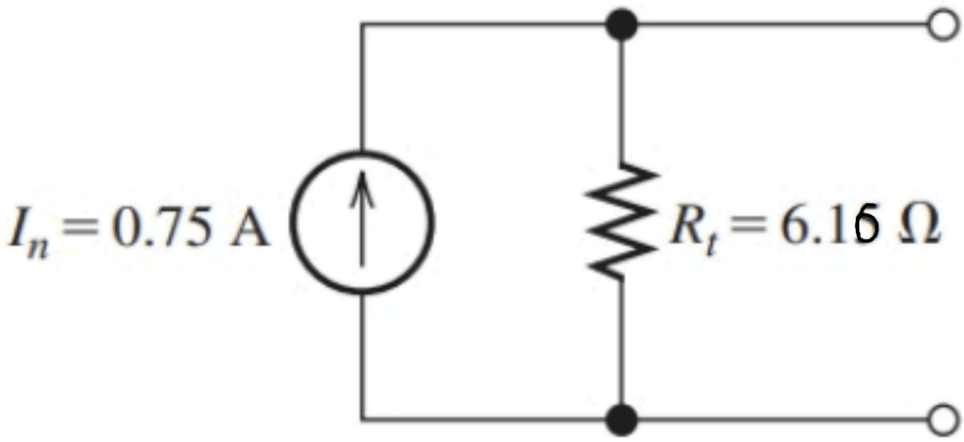
Thus, the short circuit current is only due to the 15V source, that is

$$i_{sc} = \frac{15}{20} = 0.75A$$

Thus, the Thevenin's resistance is

$$R_{th} = \frac{4.62}{0.75} = 6.16A$$

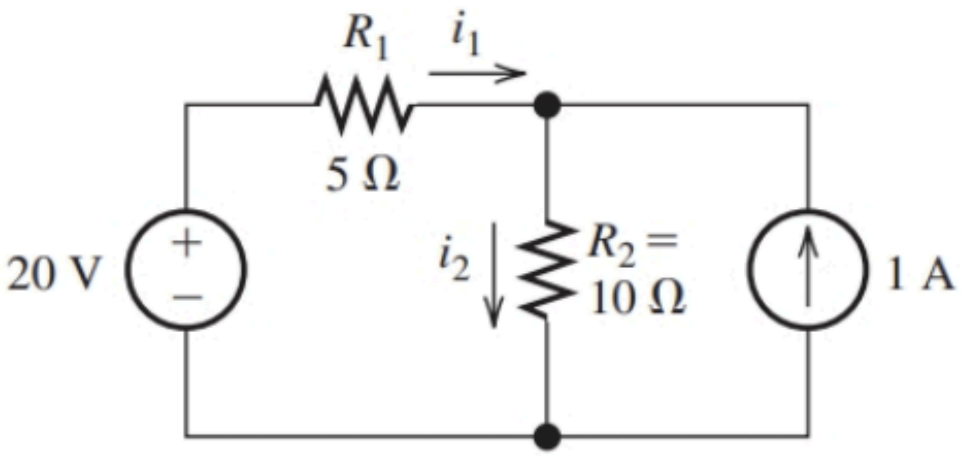
Hence, the Norton's equivalent circuit is



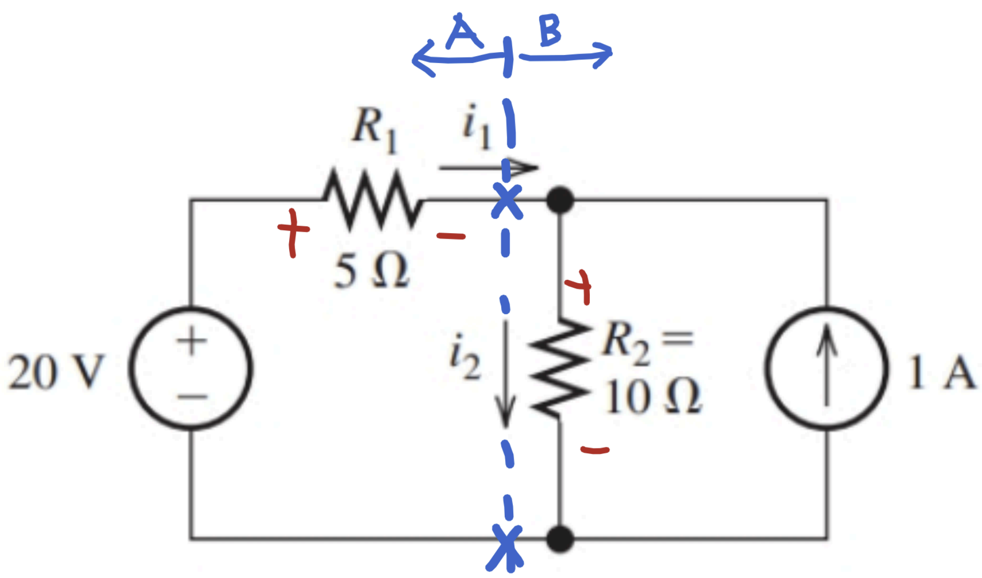

Source Transformation

Source transformation involves using Thevenin's or Norton's theorem to simplify sections of the circuit, to make circuit analysis easier. This could involve converting a section of a circuit from having a voltage source to a current source via Norton's Theorem, or vice versa with Thevenin's Theorem.

Say we had to determine i1 and i2 from the circuit shown:




We can split the circuit into two open circuits, like so:




To determine i1, we can apply Thevenin's Theorem on the RHS circuit, or circuit B, to get a voltage source and resistor in series.

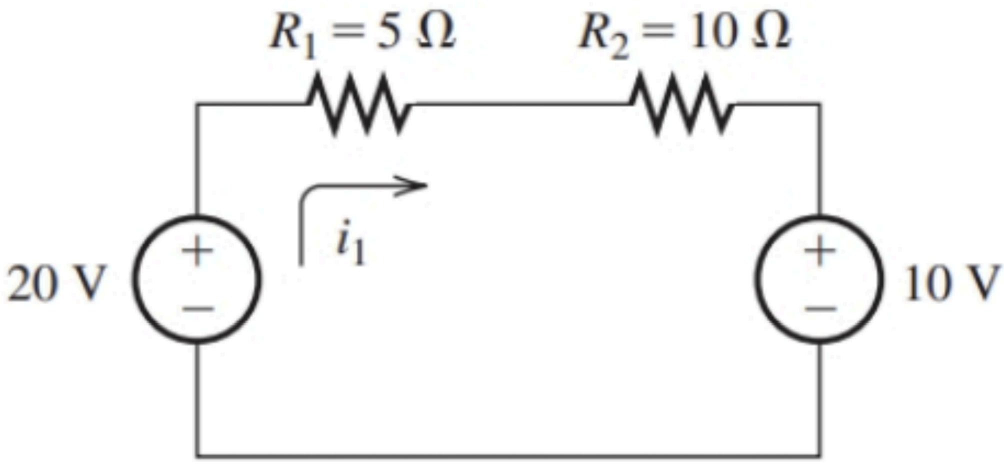
Hence,

$$R_{th} = R_2 = 10\Omega$$

and since $i_{sc} = 1A$,

$$V_{th} = 1 \times 10 = 10V$$

Thus, the circuit can be redrawn as




i1 can now be easily determined, by utilising KVL:

$$\begin{gathered}-20 + 5i_1 + 10i_1 + 10 = 0 \\ 15i_1 = 10 \\ i_1 = \frac{10}{15} = 0.667A\end{gathered}$$

Then, by applying KCL on the top node in the original circuit:

$$\begin{gathered}i_1 + 1 = i_2 \\ i_2 = 1.67A\end{gathered}$$

An alternative method is by applying Norton's Theorem on Circuit A (LHS) to get a resistor in parallel, and finding i2.

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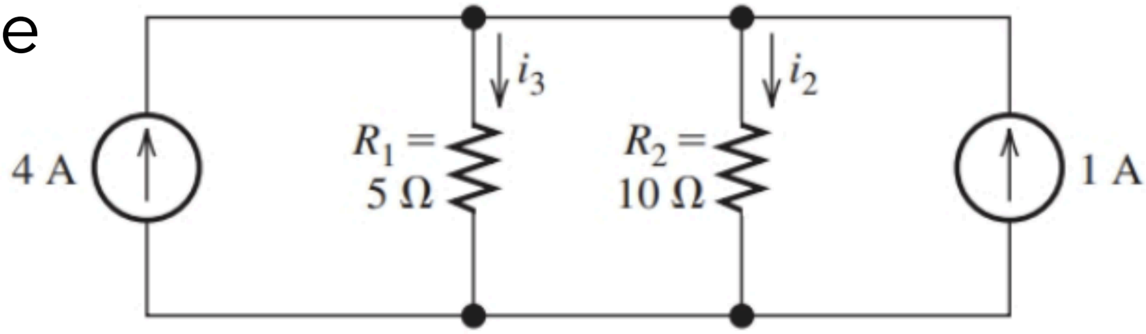



Note how the current flowing through the 5 ohm resistor is no longer i1, and is a different value.

After finding i2 by CDR, i1 can be found from the original circuit.

Principle of Superposition

Recall the Superposition theorem from PEEE1, which more formally is

$$v_p = \sum_{k=1}^{n} v_k|_p, \; i_p = \sum_{k=1}^{n} i_k|_p$$

Essentially, the total voltage or current of a component is the sum of the individual voltages or currents due to each power source.

The principle of superposition only works for linear circuits, that is circuits that operate in the linear region, meaning current and voltage are proportional and additive.

Recall the general procedure:

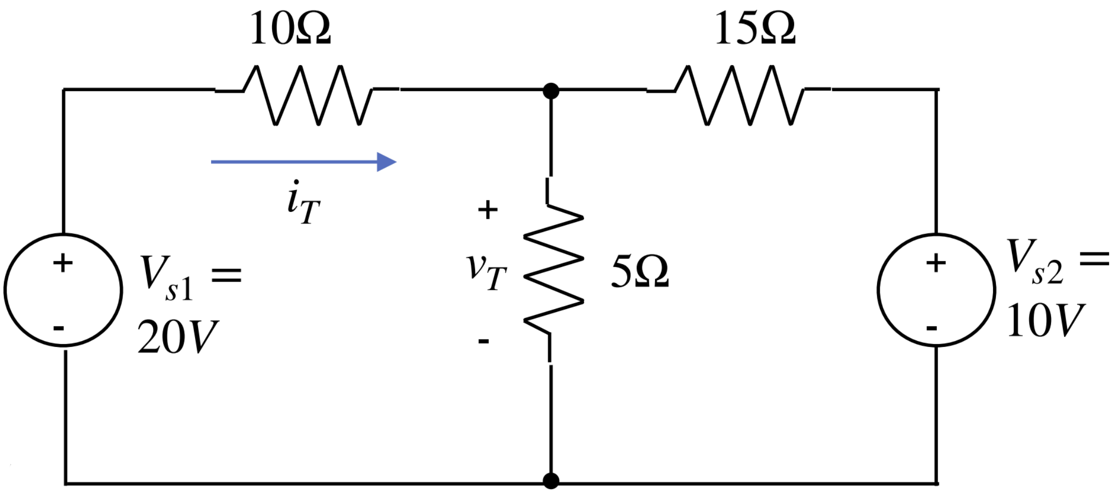
Zero the sources except source of interest

find desired value due to source of interest

repeat for number of sources

Add the values together, including signs.

Say we had to determine $i_T$ and $v_T$ from the circuit:

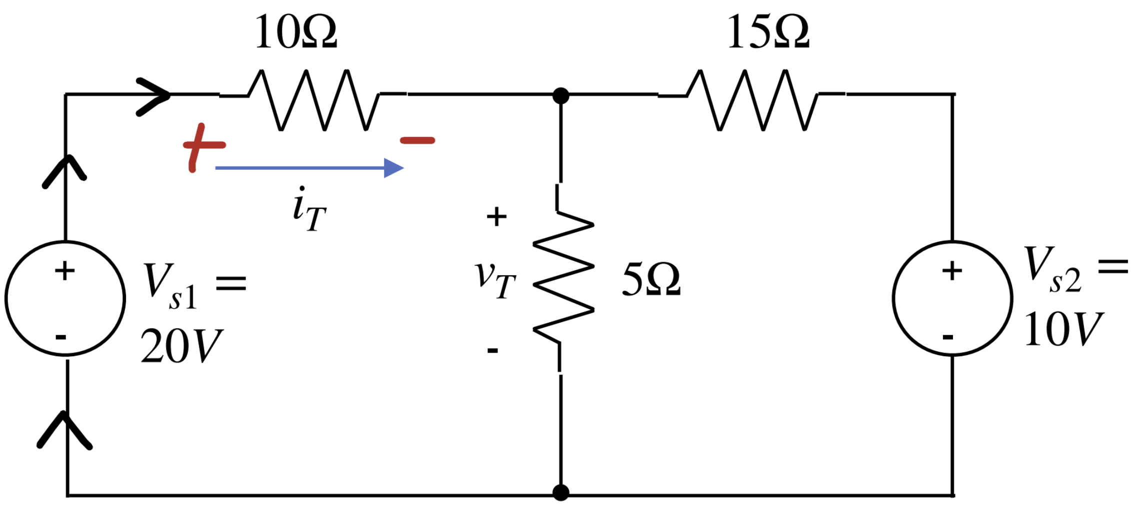



First, we will find $v_T$:

Short $V_{s2}$ to find $v_T|_{V_{s1}}$:

$$\begin{aligned} v_T|_{V_{s1}} &= \frac{5//15}{10 + 5//15} \times 20 \\ &= \frac{\frac{5\times 15}{5+15}}{10 + \frac{5\times 15}{5+15}} \times 20 \\ &= \frac{3.75}{13.75} \times 20 \\ &= 5.45V \end{aligned}$$

We can also determine $i_T|_{V_{s1}}$ from here too, by considering the source current due to voltage source 1. For this case, the direction of the current we calculate aligns with the given convention for $i_T$, thus $i_T|_{V_{s1}}$ will be positive:



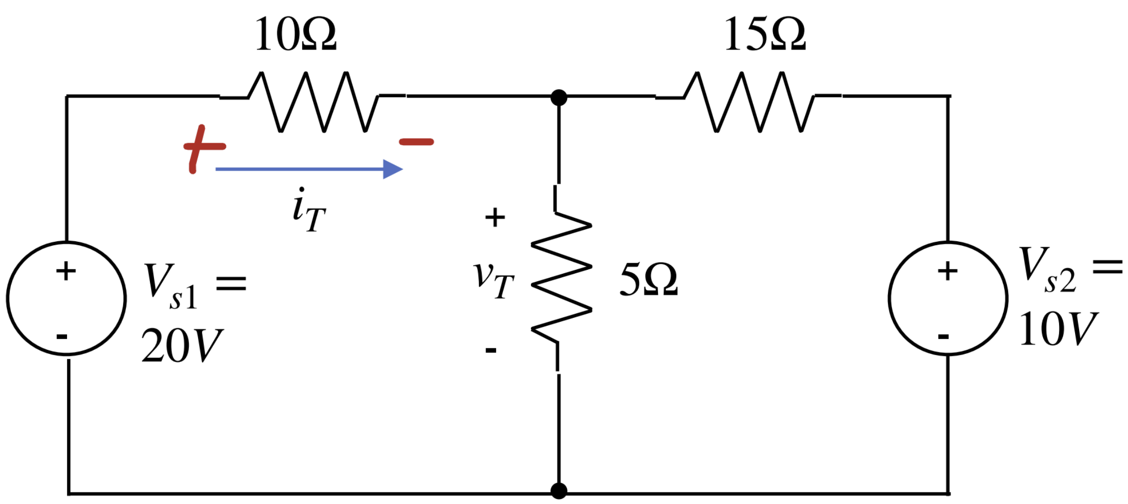
Hence,

$$
\begin{aligned}
i_T|_{V_{s1}} &= \frac{20}{10 + 5//15} \\
&= \frac{20}{13.75} = 1.45A
\end{aligned}
$$

Then, short $V_{s1}$ to find $v_T|_{V_{s2}}$ and $i_T|_{V_{s2}}$:

$$
\begin{aligned}
v_T|_{V_{s2}} &= \frac{5//10}{15 + 5//10} \times 10 \\
&= \frac{\frac{5\times10}{5+10}}{15 + \frac{5\times10}{5+10}} \times 10 \\
&= \frac{\frac{10}{3}}{15 + \frac{10}{3}} \times 10 \\
&= \frac{\frac{10}{3}}{\frac{55}{3}} \times 10 \\
&= \frac{20}{11} = 1.82V
\end{aligned}
$$

$i_T|_{V_{s2}}$ can be determined by parallel branches, however based on the direction $i_T$ was defined as, the voltage across the 5 ohm and 10 ohm resistors are opposites:




Hence,

$$
i_T|_{V_{s2}} = -\frac{\frac{20}{11}}{10} = -0.182A
$$

Thus, sum the voltages and currents:

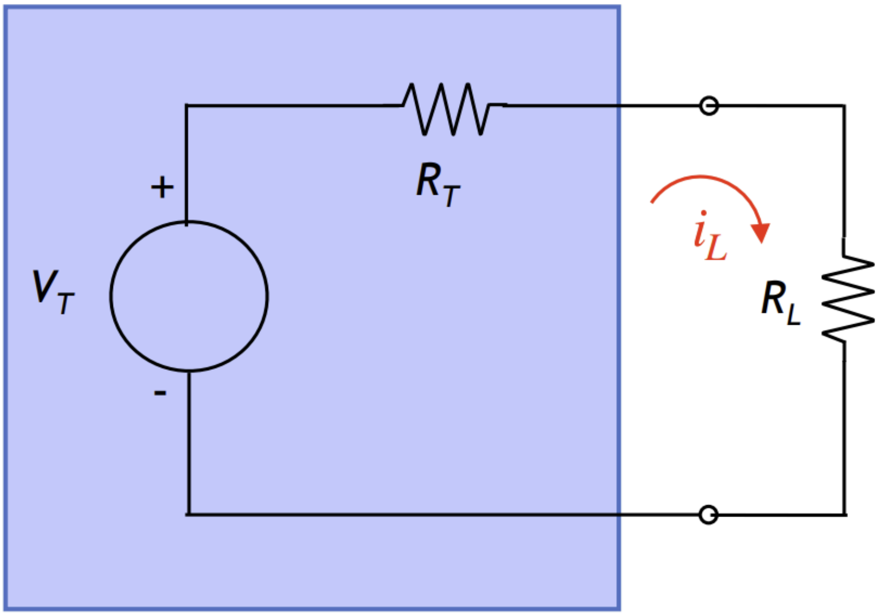
$$v_T = 5.45 + 1.82 = 7.27V$$
$$i_T = 1.45 - 0.182 = 1.268A$$

Remember to consider the sign.

## Maximum Power Transfer

Say we had a two terminal circuit and connected it to a load resistor of resistance $R_L$. What resistance will give the maximum power dissipated?

Firstly, we know that the circuit can be simplified with Thevenin's theorem




*Thévenin equivalent circuit with load $R_L$*

Hence, since the load current $i_L$ is just

$$i_L = \frac{V_T}{R_T + R_L}$$

Hence, the power dissipated is

$$P_L = i_L^2 R_L = \frac{V_T^2 R_L}{(R_T + R_L)^2}$$

Thus, to find the maximum possible $P_L$ from $R_L$, we let $\frac{dP_L}{dR_L} = 0$. First,

$$\begin{aligned}\frac{dP_L}{dR_L} &= \frac{V_T^2(R_T+R_L)^2 - 2(R_T+R_L)(V_T^2R_L)}{(R_T+R_L)^4} \\ &= \frac{V_T^2(R_T+R_L) - 2V_T^2R_L}{(R_T+R_L)^3} \\ &= \frac{V_T^2R_T + V_T^2R_L - 2V_T^2R_L}{(R_T+R_L)^3} \\ &= \frac{V_T^2R_T - V_T^2R_L}{(R_T+R_L)^3} \\ &= \frac{V_T^2(R_T-R_L)}{(R_T+R_L)^3}\end{aligned}$$

Thus, when you let P’=0,

$$\begin{aligned}V_T^2(R_T-R_L) &= 0 \\ R_L &= R_T\end{aligned}$$

Thus, the maximum power dissipated is when the resistive load is equal to the Thevenin’s resistance. Substituting back into the power expression, we find the maximum power produced is

$$\begin{aligned}P_L &= \frac{V_T^2R_L}{(R_T+R_L)^2} \\ &= \frac{V_T^2R_L}{(2R_L)^2} \\ &= \frac{V_T^2R_L}{4R_L^2} \\ &= \frac{V_T^2}{4R_L}\ [W]\end{aligned}$$

Some things to note about this maximum power value:

A maximum power does not mean maximum efficiency, as the same power is dissipates in the internal Thevenin resistance, thus half of the power is lost in the source resistance.

We use maximum power transfer in signal processing to maximise the signal produced

However, in power distribution systems, we use efficiency as a more important metric, due to the power loss said earlier.